PORTFOLIO OPTIMIZER 2.0

# Which Ideas Are Real, Useful, or Just Trading Lore?

An evidence-led feasibility memo covering overnight returns, statistical arbitrage, pairs, derivatives, Kelly sizing, Monte Carlo, ICIR research loops, intraday rules, alternative data, SMRs, and the proposed open-source tools.

**Bottom line: keep the existing safety-first architecture. Add a stronger experiment-selection layer and a small number of predeclared challengers. Do not confuse a documented anomaly, a useful library, or an attractive story with a deployable edge.**

Prepared for project discussion | Full working analysis | Research-only, not personal financial advice

# 1. Executive decision

> **The most valuable change is not another strategy. It is a stricter strategy factory: predeclare the hypothesis, choose the metric that matches the signal, retain every failed trial, use nested chronological validation, and protect one final holdout from feedback. The project already does much of this; the next step is to formalize ICIR, trial-aware performance correction, and one-shot promotion gates.**

| Idea | Is the claim true? | Project fit | Decision |
|---|---|---|---|
| Overnight effect | Historically yes; recently unstable | Medium | Diagnostic challenger |
| Stat arb / pairs | Real but probabilistic | Medium | Do not retune failed ETF test |
| Indonesia pairs | Possible inefficiency; shorting is binding | Low now | Research only / wait for borrow |
| First 5-min candle / 12 EMA | Exact rule is unproven | Low in current architecture | Separate intraday sandbox |
| Options disagreement | Informative attention signal, not free alpha | Medium-high | Build feature layer first |
| ICIR + OOS feedback loop | Sound with holdout discipline | Very high | Implement next |
| Monte Carlo | Useful for falsification, never proof | Already strong | Extend, do not replace |
| Kelly criterion | Useful after edge validation | High later | Fractional, constrained challenger |
| Ranked Asset Allocation | Real paper; backtest not independently proven | Medium | Replicate post-2017 |
| NuScale / SMR | Theme is real; economics remain binary | Low as optimizer alpha | Watchlist / capped satellite |
| Politician / influencer copy | Late, noisy, selection-biased | Low | Use only as delayed feature |
| Scrapling / OpenBB | Useful data tools with caveats | Medium | Isolated adapters |
| Kronos | Real foundation model, not a quant-firm clone | Medium | Quarantined challenger |
| NautilusTrader / DASReplay | Useful for execution work | Low until intraday | Defer |
| Everything Claude Code | Developer workflow bundle | Indirect | Do not install wholesale |

## Recommended priority order

- 1. Formalize the experiment loop and one-shot promotion gate.
- 2. Add options-derived features as an information layer, not an execution product.
- 3. Replicate RAAM and test a clean overnight decomposition with post-publication data.

- 4. Trial Kronos only as a challenger and OpenBB/Scrapling only as guarded adapters; defer intraday engines, Indonesia long-short pairs, and derivative trading until their data, borrow, and accounting gates exist.

# 2. What the project already has

Portfolio Optimizer 2.0 is not a blank backtester. It is explicitly a safety-first research and simulation platform. Third-party code is restricted to replaceable research inputs or adapters; platform-owned data vintages, execution, accounting, risk, and promotion gates remain authoritative. The implemented flow already maps closely to a professional research stack: immutable inputs -> timestamped signals -> portfolio construction -> independent risk gate -> replay/paper execution -> audit trail and dashboard. [P1][P2]

| Capability | Current evidence | Implication for these ideas |
|---|---|---|
| Pairs / stat arb | Independent Engle-Granger engine, cost/borrow accounting, controls | The prior failure is meaningful; do not assume a coding bug. |
| Factor IC | Cross-sectional Spearman RankIC, chronological folds, controls | ICIR is a small extension, not a new platform. |
| Monte Carlo | IID, block, mean-haircut and forced-crash paths with calibration | Use it as risk falsification, not forecast alpha. |
| Research loop | Deterministic experiment IDs, immutable snapshots, walk-forward selection | Add a formal trial ledger and final untouched gate. |
| Indonesia | Point-in-time IDX80 membership and official fundamental archives | Promising research lane, but data and trading constraints remain. |
| Intraday / derivatives | No minute schema, Greeks, margin, assignment or variation margin | Requires a separate architecture lane. |

> **Current truth: the portfolio has only 1 of the required 52 untouched forward weeks, and no strategy is approved for real-money execution. The correct posture is 'research-grade candidate,' not 'proven system.' [P10]**

A focused pure-Python audit of the pair, Monte Carlo, research-lab, evaluation, and data-vintage components passed 27 tests. A broader Indonesia test run was blocked by a local pandas import hang, and pytest is not installed in the project environment. That is a runtime-maintenance issue, not strategy evidence.

# 3. 'The market makes most of its money while closed'

**TRUE HISTORICALLY; NOT A DURABLE FREE TRADE**

The statement describes a return decomposition: close-to-open returns have explained a surprisingly large share of long-run U.S. equity gains in several samples, while open-to-close returns were much weaker. Research also links overnight drift to market-maker inventory and overseas market openings. [S1][S2]

But the actionable version is weaker than the slogan. In July 2026, the New York Fed reported that the well-known 2:00-3:00 a.m. Eastern drift averaged close to zero after 2021. That is exactly what we should expect from public anomalies: they can migrate, decay, or be conditional on market structure. [S3]

## How we can use it:

- First, add an attribution report splitting every strategy and benchmark into close-to-next-open and open-to-close returns. This tells us whether our apparent alpha is really gap beta.
- Only then test an overnight challenger: market-on-close entry, next market-on-open exit, auction-aware costs, dividends and corporate actions, earnings-event strata, gap-tail risk, and distinct pre- and post-publication eras.
- Reject a rule that works only in the old sample, only at frictionless closing/opening marks, or only because the universe omits delisted stocks.

> **Use the information as a diagnostic first. A stop order cannot protect against a gap while the cash market is closed, and buying the close / selling the open creates concentrated auction execution risk.**

# 4. Statistical arbitrage, 'mispricing,' and the failed pair system

**REAL FIELD; NO ALWAYS-WIN STRATEGY**

Statistical arbitrage is a family of positive-expected-value bets, not arbitrage in the riskless textbook sense. Classical pairs research found attractive historical returns, while later work showed substantial decay after costs and after 2002. Structural breaks, hedge-ratio drift, borrow, impact, and multiple testing are core failure modes. [S4][S5]

The project's test was already rigorous. It rebuilt a broken public sample rather than trusting it, used 504-day formation windows, Engle-Granger tests, Benjamini-Hochberg false-discovery control, disjoint pairs, fixed z-score rules, relationship-break stops, next-close realization, full traded-notional costs, borrow fees, and random/inverted/stale controls. Across 126 formations, the gross sleeve earned only 0.047% CAGR with a 0.031 Sharpe. At 50 bps plus 3% borrow, it lost 1.518% annually, produced a -0.751 Sharpe and -30.38% drawdown, and worsened the core portfolio. [P3][P4]

> **Conclusion: nothing here points to 'we implemented pairs wrong.' The tested daily ETF mean-reversion hypothesis had no economic edge. Re-optimizing thresholds on the same history would turn a valid failure into overfitting.**

### A genuinely new pairs experiment must change the economic hypothesis, not just parameters:

- Prefilter economic substitutes within the same industry or share class; avoid pure correlation fishing.
- Estimate hedge ratios only in the formation window; monitor beta, sector, factor and liquidity residuals.
- Use fixed entry, exit, relationship-break and time-stop rules, then walk-forward reselection.
- Require actual borrow availability, borrow fees, locate constraints, bid/ask data and suspension handling.
- Use a new untouched time period or market. Do not call a long-only spread simulation 'pairs trading.'

# 5. Can Indonesia make pairs work better?

**INTERESTING MARKET; CURRENTLY CONSTRAINED EXPRESSION**

Less-refined price discovery can create opportunities, but it also creates higher spreads, thinner depth, stale prices, suspensions, price limits, corporate-action complexity, and greater implementation risk. Inefficiency is not the same as accessible net alpha.

The binding issue is the short leg. OJK's 2024 framework regulates short-sale financing, but OJK continued to report postponement of its implementation in 2025 and 2026. A retail or small-institution market-neutral IDX pair may therefore be unavailable or severely constrained. [S6][S7]

### Recommended Indonesia path

| Stage | Test | Gate |
|---|---|---|
| 1. Data | Point-in-time IDX80, inactive securities, corporate actions, daily bid/ask or conservative spread model | No performance claim until coverage and total-return audit pass |
| 2. Long-only relative value | Sector-neutral residual reversal or economically matched ranking | Must beat simple momentum/low-vol after 25-150 bps stress |
| 3. Shadow pairs | Record hypothetical long/short signals and actual short eligibility/borrow | At least 12 months feasible expression evidence |
| 4. True pairs | Only if broker, borrow, regulations and settlement are operational | New untouched gate; no backfilling missed opportunities |

The current Indonesian momentum/fundamental work already found that realistic costs and inactive-security repair materially weaken results. That is a reason to be more conservative in Indonesia, not less. [P5]

# 6. First 5-minute candle vs 12 EMA with an ATR trail

**TESTABLE; EXACT RULE HAS NO DEMONSTRATED EDGE**

The rule is coherent as a trend-following hypothesis: use early-session direction as the entry, then let volatility define risk and allow winners to run. Academic evidence supports related but different opening-range and intraday-continuation effects; it does not validate this exact 5-minute/12-EMA combination. [S8]

Before testing, freeze every ambiguous detail: regular-hours-only or premarket EMA; whether prior-day bars seed the EMA; signal timestamp at 9:35 ET; entry at the next executable quote; ATR lookback and multiplier; whether trailing stops update from completed bars; short-sale assumptions; forced exit time; and earnings/macro-news treatment.

### Required experiment design

- Use one-minute quotes or NBBO-aware bars, opening-auction conditions, spreads, latency and next-quote fills. A same-candle close fill is lookahead.
- Compare against the sign of the first candle alone, EMA alone, an opening-range breakout, and buy-and-hold intraday.
- Perturb EMA 8-20 and stop 1-4 ATR. If only exactly 12 EMA and one stop setting work, reject it.
- Separate SPY, QQQ and ES from single stocks; stratify by volatility, gap size, earnings and market regime.
- Build this in an isolated intraday lane. The current platform aggregates daily data to weekly decisions and cannot test it faithfully. [P6]

## 7. Derivatives: what they are and how they help

**HIGH RESEARCH VALUE; TRADING LAYER COMES LATER**

A derivative is a contract whose value depends on an underlying asset or rate. Futures create an obligation to transact later; options give the buyer a right; swaps exchange defined cash flows. They can hedge, create leverage, express volatility, and reveal market-implied pricing. They also add nonlinear risk, margin, expiry, assignment, settlement, liquidity and model assumptions. [S9][S10]

### Best first use for this project: information, not execution

| Feature | What it measures | Main trap |
|---|---|---|
| ATM implied volatility | Option-implied magnitude of future movement | Not direction; includes volatility risk premium |
| Skew | Relative price of downside vs upside protection | Crash insurance demand and dealer positioning |
| Term structure | Near-term vs longer-term uncertainty | Event calendars and stale/misaligned expiries |
| Put/call volume and OI | Activity and positioning proxies | Opening/closing and buyer/seller direction are unknown |
| Option-implied stock price | Put-call-parity consistency | Quotes, dividends, early exercise and microstructure |

The Instagram claim - compare stock price with what people are betting happens next and investigate disagreement - is a sensible attention filter. It is not a prediction by itself. A major study of option-implied stock prices found that option quotes adjusted to remove parity disagreement while stock quotes behaved normally; there was no economically meaningful stock-price discovery edge. [S11]

> **Build a ranked research feature: price trend or fundamental signal vs IV/skew/term-structure surprise. Test whether disagreement improves an independently valid signal. Do not trade every disagreement.**

The current execution engine does not model option multipliers, Greeks, margin, early exercise, assignment, expiry settlement or futures variation margin. Actual derivative trading is therefore a material architecture project, not a plug-in. [P7]

# 8. Kelly criterion

**USE FOR SIZING VALIDATED EDGES ONLY**

Kelly maximizes expected logarithmic wealth, or long-run geometric growth, when the probability distribution is known. It does not create an edge. Estimation error is the practical danger: full Kelly applied to an optimistic mean or understated tail can produce extreme leverage and intolerable drawdowns. [S12][S13]

## Recommended project implementation:

- Estimate a joint scenario distribution for validated strategy sleeves, not a win-rate formula for isolated trades.
- Shrink expected returns aggressively toward zero and inflate covariance/tail estimates.
- Compare quarter-Kelly and half-Kelly with inverse-volatility, volatility targeting and existing minimum-variance allocation.
- Apply hard caps on asset weight, sleeve weight, gross exposure, turnover, liquidity and drawdown probability.
- If the constrained Kelly solution exceeds the cap, the cap is the true size. If the edge is not decision-grade, size is zero.

A risk-constrained Kelly challenger is viable after the forward-evidence clock and strategy calibration mature. It is not the next strategy discovery project.

# 9. The quant research loop, ICIR, and the OOS gate

**HIGHEST-PRIORITY IMPROVEMENT**

The described loop is broadly how disciplined systematic research works: generate a falsifiable hypothesis, backtest under a fixed protocol, diagnose the failure, revise, and compare against an explicit hurdle. The danger is that every observed validation result becomes training information. Repeating the loop against the same 'out-of-sample' period eventually makes it in-sample.

## For a cross-sectional ranking signal at date t:

> **IC(t) = Spearman correlation(signal ranks at t, future return ranks). ICIR = mean(IC) / standard deviation(IC), with the annualization convention stated. ICIR measures consistency of ranking skill; it is not a universal strategy score.**

The project already computes mean RankIC and IC standard deviation and has chronological dev/validation/test partitions, inversion, staleness and permutation controls. It does not currently expose ICIR by name. Adding it is easy; using it correctly is the important part. [P8]

| Layer | Purpose | Can feedback enter? |
|---|---|---|
| Research / train | Generate and fit hypotheses | Yes |
| Nested validation | Choose model family and diagnose | Yes; then it becomes training knowledge |
| Locked test | Compare the frozen candidate once | No tuning after inspection |
| Untouched forward | Accumulate evidence after freeze | No backfill, no mutation |

Do not optimize every strategy to ICIR. Use RankIC/ICIR for cross-sectional stock selection; cost-adjusted Sharpe/Calmar and tail loss for portfolio timing; execution shortfall and fill quality for intraday systems; and calibration/error metrics for forecasts. Promotion should combine predictive stability, realized net returns, tail risk, turnover/capacity, multiple-testing correction and operational readiness.

Add White's Reality Check or a comparable familywise test, Deflated Sharpe Ratio, and Probability of Backtest Overfitting/CSCV to the experiment ledger. These do not prove profitability; they make it harder for the best-looking trial to be a lucky winner. [S14][S15][S16]

# 10. Monte Carlo and other ways to test survival

**ALREADY USEFUL; BROADEN THE FALSIFICATION SUITE**

Monte Carlo cannot prove that a strategy 'will work.' It can show how often it fails under assumptions and expose sequence, regime, tail, cost and concentration fragility. The project already does more than naive Gaussian paths: 13-week circular blocks, positive-mean haircut, forced historical crash starts, tail probabilities, expected shortfall and rolling calibration. [P9]

| Test | Question answered |
|---|---|
| Moving/stationary block bootstrap | Does the result survive serial dependence and alternative sequences? |
| Mean haircut / Bayesian shrinkage | What if the estimated edge is overstated? |
| Regime and correlation shock | What happens when diversification disappears? |
| Cost / missed-fill / delay stress | Is the edge larger than implementation error? |
| Remove top trades / issuers | Is performance concentrated in a few accidents? |
| Parameter neighborhood | Does the economic idea work near the chosen setting? |
| Cross-market replication | Is it a general mechanism or one dataset? |
| Negative and placebo controls | Would random, inverted, stale or nonsense features also pass? |
| Untouched forward clock | Does a frozen rule survive new information? |

**The existing five-year block model estimated only a 3.5% probability of ending below starting wealth, but a positive-mean haircut raised it to 18.8%; forced-crash paths ended below starting wealth 33.8% of the time. The assumption is the result. [P9]**

# 11. Ranked Asset Allocation Model (RAAM)

**REPLICATE, THEN JUDGE**

RAAM is a real 2018 Charles H. Dow Award paper by Gioele Giordano. It starts from the 7Twelve multi-asset ETF universe, ranks non-cash assets using absolute momentum, volatility, average relative correlation and an ATR trend/breakout component, selects five, and reallocates monthly. The paper reports very strong gross 2004-2017 results. [S17]

Why it is not ready to adopt: transaction and management fees were excluded; parts of the indicator design are proprietary or opaque; interpolation and ETF inception/proxy choices require audit; and there is no untouched post-2017 test in the paper.

Project fit is good because the current stack already has momentum, volatility, correlation, trend, inverse-volatility allocation, costs and point-in-time controls. That also means RAAM may be a recombination of existing exposures rather than a new independent sleeve.

## Replication gate

- Reconstruct the 12-asset universe with total-return data and an explicit pre-inception proxy policy.
- Implement the paper literally before changing anything; record unclear choices as branches, not hidden judgment.
- Use July 2004-November 2017 only for replication; reserve December 2017 onward as the decisive test.
- Compare with equal-weight 7Twelve, SPY, simple absolute momentum, risk parity and the existing optimizer after costs.
- Run lookback, weight and universe neighborhoods plus a leave-one-asset-class-out test.

# 12. NuScale and small modular reactors

**REAL TECHNOLOGY MILESTONE; SPECULATIVE EQUITY**

SMRs are a legitimate long-duration energy theme. NuScale's US460 received NRC standard design approval in May 2025, allowing the design to be referenced in applications to build and operate plants. Approval establishes a regulatory design milestone, not project economics or a construction order. [S18]

NuScale's 2025 Form 10-K shows $31.5 million of revenue, a $664.5 million net loss, $459.6 million of operating cash use, and substantial liquidity - but also about $1.30 billion of net financing inflow from issuing 57.1 million Class A shares. The first UAMPS project was terminated in 2023. Commercial success remains milestone- and dilution-dependent. [S19][S20]

| Evidence window | What would improve the thesis | What would kill it |
|---|---|---|
| Customer | Binding contract with credible offtaker | Non-binding announcements or cancellations |
| Financing | Fully financed project and cost-sharing clarity | Repeated equity issuance without contracted cash flow |
| Licensing | Site/construction/operating approvals | Material redesign or schedule slippage |
| Economics | FOAK cost and delivery schedule validated | Cost escalation makes power uncompetitive |
| Execution | Module manufacturing and first delivery | Supply-chain or quality failure |

For this project, NuScale is better treated as a monitored high-volatility thematic exposure or event-driven watchlist than as a central systematic strategy. If included, use scenario sizing and a hard cap; Kelly cannot rescue an uncertain distribution.

# 13. Copying politicians or influencers

**DO NOT COPY; TEST AS DELAYED ALTERNATIVE DATA**

Congressional disclosures are not live trade feeds. House rules permit reporting by the earlier of 30 days after awareness or 45 days after the transaction, and values are disclosed in ranges. Research suggesting historical congressional alpha is much weaker after the 2012 STOCK Act. [S21][S22]

A fair backtest must trade only after the actual public filing timestamp, include all members rather than selecting a famous winner after the fact, map spouse/options transactions correctly, model amount ranges, and control for market, sector, size and momentum. It should compare against placebo politicians and shuffled filing dates.

Influencers are worse as a direct signal: posts may be promotional, deleted, delayed, sponsored, or capable of moving illiquid prices after followers react. The SEC explicitly warns about social-media stock scams. [S23]

> **Useful version: archive timestamped filings/posts and turn them into slow sentiment, crowding or event features. Invalid version: automatic replication orders.**

# 14. Repositories and tools: what each actually does

| Tool | What it is | Verdict for this project |
|---|---|---|
| Scrapling | BSD-3 Python scraping/crawling framework; not merely an extension | Guarded adapter for public pages with no API. Archive raw HTML, obey terms/robots, validate drift. Never replace official/vendor truth. [S24] |
| Kronos | MIT OHLCV candlestick foundation model with open small/base weights | Quarantined challenger only. Benchmark against simple ranks with nested/embargoed OOS, multiple seeds and costs. Not a clone of quant firms. [S25] |
| OpenBB | AGPL open data-integration platform; it does not provide all data | Trial one noncritical provider adapter. Preserve project provenance/vintages; review AGPL and each provider's terms. [S26] |
| NautilusTrader | LGPL Rust/Python event-driven research/live engine | Strong future candidate for tick/order-book or broker parity; overkill for weekly optimizer and no assumed IDX adapter. [S27] |
| Everything Claude Code / ECC | MIT developer-agent workflow/config bundle | No direct investment value. Do not install wholesale; audit and cherry-pick isolated workflow ideas. [S28] |
| DASReplay | US-market manual replay/training product; current beta | Potential qualitative execution rehearsal for the 5-minute rule. Not a batch backtest API, not Indonesia, not proof. [S29] |
| Trading Systems repo | Identity ambiguous | No installation or assessment until the exact URL/owner is provided. |

## Integration rule

Every third-party tool stays behind a narrow adapter. Pin the commit/version and license, run in isolation, compare identical inputs with a platform-owned baseline, preserve raw inputs and outputs, and prohibit direct access to canonical portfolio state or execution. This is already the project's stated architecture and should not change for a popular repository. [P2]

# 15. Concrete experiment backlog

| Priority | Experiment | Primary metric | Promotion gate |
|---|---|---|---|
| P0 | Research governance v2: ICIR, trial ledger, DSR/PBO or familywise correction, locked test + forward gate | Metric depends on strategy class | Cannot promote if final holdout was queried during iteration |
| P1 | Options-information layer: IV, skew, term structure and parity disagreement joined point-in-time | Incremental RankIC/ICIR and net portfolio ablation | Positive familywise OOS increment after data and turnover costs |
| P1 | RAAM literal replication and post-2017 test | Net Sharpe/Calmar and benchmark-relative return | Beats simple momentum/risk parity without concentration or proxy dependence |
| P1 | Overnight attribution + challenger | Net close-open return, ES, gap drawdown | Survives post-2021 era and auction costs |
| P2 | Kronos OHLCV challenger | Incremental RankIC/ICIR vs simple price factors | Multiple seeds, no pretraining-overlap concern, net OOS improvement |
| P2 | Indonesia long-only relative-value challenger | Net alpha vs matched long-only baseline | PIT universe, inactive securities, 150 bps stress, no short claim |
| P3 | First-5-minute/EMA/ATR sandbox | Net intraday return and execution shortfall | Broad parameter plateau; next-quote fills; regime stability |
| P3 | Political/social disclosure feature | Incremental OOS signal value after public timestamp | All-person universe, delay, placebo and multiple-testing controls |
| Defer | True IDX pairs, derivative execution, Nautilus integration | Operational readiness | Borrow/data/accounting/execution prerequisites cleared |

## Standard challenger contract

- Write the economic mechanism and falsifier before code.
- Freeze universe, signal timing, data availability, costs, borrow, fill model and benchmark.
- Choose one primary metric and a small set of risk/implementation gates; do not change them after seeing results.
- Run negative controls, parameter neighborhoods, delayed/stale inputs and missing-data stress.
- Use nested chronological selection, then a locked test once; preserve every trial in the registry.
- If it passes, create a new frozen version and start an untouched forward clock. A passed backtest is a candidate, not an approval.

> **Decision ask: approve P0 and the three P1 research tracks. Keep all other ideas in a research queue until their prerequisite data and market-access gates are explicit.**

# 16. Final answers in plain language

## Can we exploit the market being closed?

Maybe as a conditional overnight sleeve, but first use it to diagnose where returns occur. The famous drift has weakened materially since 2021.

## Can we find a strategy that always wins?

No. If a price inconsistency were riskless, executable and persistent, competition would remove it. What remains is uncertain edge with model, funding, execution and tail risk.

## Did our pair system fail because we did it wrong?

The evidence says mostly no. The public sample was broken, but our independent repair was causal and rigorous. The economic edge vanished before costs and became clearly negative after costs.

## Could pairs work in Indonesia?

Possibly as research, but genuine market-neutral implementation is currently limited by short-sale financing and local microstructure. Start with long-only relative value and a short-eligibility shadow record.

## Should we use more math?

Yes, to expose uncertainty and fragility. More math does not turn an unstable assumption into truth.

## Should we add the repositories?

Only behind isolated adapters or challenger harnesses. None should replace the project's data, accounting, risk or promotion controls.

## What is the best idea here?

The research loop with correct metric ownership, trial accounting and a genuinely untouched gate. It improves every future strategy search.

## Source posture

This memo combines direct inspection of the local project with academic papers, official regulators/exchanges, official repository documentation and the latest available company filing. The Instagram reel itself could not be fetched due to Instagram throttling; its discussion is limited to the user's paraphrase and the closest identifiable options-pricing literature. The generic 'Trading Systems' repository could not be identified without a URL.

# Appendix A. External sources


S1 Cooper, Cliff and Gulen, Return Differences between Trading and Non-Trading Hours
https://papers.ssrn.com/sol3/Delivery.cfm/SSRN_ID1266293_code265947.pdf?abstractid=1004081

S2 New York Fed, The Overnight Drift
https://www.newyorkfed.org/medialibrary/media/research/staff_reports/sr917.pdf

S3 New York Fed, The Disappearing Overnight Drift, July 2026
https://libertystreeteconomics.newyorkfed.org/2026/07/the-disappearing-overnight-drift/

S4 Gatev, Goetzmann and Rouwenhorst, Pairs Trading
https://www.nber.org/papers/w7032

S5 Do and Faff, Are Pairs Trading Profits Robust to Trading Costs?
https://onlinelibrary.wiley.com/doi/10.1111/j.1475-6803.2012.01317.x

S6 OJK Regulation 6/2024, Margin and Short Selling
https://ojk.go.id/id/regulasi/Pages/Pembiayaan-Transaksi-Efek-Perusahaan-Efek-Bagi-Nasabah-dan-Transaksi-Short-Selling-Oleh-Perusahaan-Efek-POJK-6-Tahun-2024.aspx

S7 OJK March 2026 market-stability update
https://www.ojk.go.id/en/berita-dan-kegiatan/siaran-pers/Pages/March-2026-Board-of-Commissioners-Meeting-Financial-Services-Sector-Stability-Remains-Maintained.aspx

S8 Gao, Han, Li and Zhou, Market Intraday Momentum
https://www.sciencedirect.com/science/article/pii/S0304405X18301351

S9 CFTC Futures Market Basics
https://www.cftc.gov/LearnAndProtect/EducationCenter/FuturesMarketBasics/index2.htm

S10 OCC Characteristics and Risks of Standardized Options
https://www.theocc.com/company-information/documents-and-archives/options-disclosure-document

S11 Muravyev, Pearson and Broussard, Is There Price Discovery in Equity Options?
https://www.sciencedirect.com/science/article/pii/S0304405X12001882

S12 Kelly, A New Interpretation of Information Rate
https://onlinelibrary.wiley.com/doi/abs/10.1002/j.1538-7305.1956.tb03809.x

S13 Busseti, Ryu and Boyd, Risk-Constrained Kelly Gambling
https://stanford.edu/~boyd/papers/kelly.html

S14 White, A Reality Check for Data Snooping
https://onlinelibrary.wiley.com/doi/pdf/10.1111/1468-0262.00152

S15 Bailey and Lopez de Prado, Deflated Sharpe Ratio
https://papers.ssrn.com/sol3/papers.cfm?abstract_id=2460551

S16 Bailey et al., Probability of Backtest Overfitting
https://carmamaths.org/resources/jon/backtest2.pdf

S17 Giordano, Ranked Asset Allocation Model
https://cmtassociation.org/wp-content/uploads/2020/03/JOTA-2020-Web-Version.pdf

S18 U.S. NRC, NuScale US460 Standard Design Approval
https://www.nrc.gov/node/2151866

S19 NuScale 2025 Form 10-K
https://www.sec.gov/Archives/edgar/data/1822966/000182296626000018/smr-20251231.htm

S20 UAMPS Carbon Free Power Project termination
https://www.uamps.com/carbon-free

S21 House Ethics, Financial Disclosure and PTR timing
https://ethics.house.gov/financial-disclosure/

S22 Huang and Xuan, congressional trading before and after the STOCK Act
https://business.gwu.edu/sites/g/files/zaxdzs5326/files/HuangXuan_STOCKAct.pdf

S23 SEC Investor.gov, Social Media Stock Scams
https://www.investor.gov/introduction-investing/general-resources/news-alerts/alerts-bulletins/social-media-stock-scams

S24 Scrapling official repository and BSD-3 license
https://github.com/D4Vinci/Scrapling

S25 Kronos official repository and paper
https://github.com/shiyu-coder/Kronos

S26 OpenBB official repository and provider documentation
https://github.com/OpenBB-finance/OpenBB

S27 NautilusTrader official repository
https://github.com/nautechsystems/nautilus_trader

S28 ECC (formerly Everything Claude Code) official repository
https://github.com/affaan-m/ECC

S29 DASReplay official site and beta product description
https://dasreplay.com/

# Appendix B. Internal project evidence

P1 2.0/README.md:3-21, 75-95 - platform scope and implemented strategy/risk components

P2 2.0/docs/architecture.md:3-58 - third-party boundary, data flow, experiment and forward-evidence principles

P3 2.0/PROJECT_HISTORY.md:1513-1556 - pair repository defect review and independent rebuild

P4 2.0/evidence/etf_pairs_batch_21/report.md - pair performance and rejection

P5 2.0/docs/INDONESIA_EQUITY_RESEARCH_V1.md:13-60, 170-203, 325-349 - universe, data gates, repaired results and costs

P6 2.0/src/systematic_trader/research_lab.py:15-28 and docs/architecture.md:34-38 - weekly/monthly data and decision cadence

P7 2.0/src/systematic_trader/execution.py:36-218 - current generic cash/order/fill model

P8 2.0/src/systematic_trader/factor_ic_protocol.py:25-61 and PROJECT_HISTORY.md:2006-2039 - RankIC implementation and gates

P9 2.0/src/systematic_trader/monte_carlo_risk.py:23-130 and evidence/monte_carlo_risk_batch_28/report.md - simulation methods and calibrated risk results

P10 2.0/research_registry/portfolio_candidates.json:66-100 - 1/52 forward weeks and missing final gates

> **This memo does not authorize trading, brokerage connectivity, or investment in any named security. It recommends research priorities and rejection gates.**